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(9A) Abstract

Online distance learning is a rapidly growing trend. Teachers around the world are required to shift their classes to an online format in a bid to keep pace with the demand. This true-experimental study was conducted to find out if the electronic modules and printed modules can help improve the Research Performance of third-year Industrial Technology students, for the school year 2021-2022. The respondents of this study were fifteen (15) randomly selected third-year Food Technology students for the experimental group and another fifteen (15) randomly selected students for the control group. Data in this study were gathered using a researcher-made Research performance test which had undergone validation and item analysis by a panel of experts in the field of Research. The major findings of the study revealed that the Research performance of the students during the pretest of the control and experimental group were both interpreted as “fair”. However, the post-test of the experimental group using the electronic modules, has a verbal interpretation of “Very high”, while the control group using the printed modules is “High”. There is no significant difference in the pretest scores on the Research performance both for experimental and control groups. However, a significant difference existed in the Research performance of the students in the pretest and posttest of the control group and experimental group. There is a significant difference in the post-test scores on the Research performance in favor of the experimental group using the electronic modules. Finally, it was found that electronic modules have a large effect on the Research performance of students.

(9B) Keywords: True Experimental Design, Pretest-Posttest, Online Distance Learning.

(10) Introduction

The novel coronavirus, commonly known as Covid-19, has been declared a pandemic by the World Health Organization (2020) as a current threat to humankind. This dire situation has successfully caused the global shutdown, including businesses, sports activities, and schools resulting in massive crisis-response migration of institutions to online learning as the educational platform.

To control the infection, physical distancing is implemented because it prevents the virus from further spreading and help flatten the curve. The policy of compulsory physical distancing has been implemented in many countries, including the Philippines, resulting in nationwide school and university closures.

In accordance with this policy, public and private academic institutions are compelled to make appropriate and timely modifications to continue delivering education and sustaining student academic progress. The teaching and learning activities were immediately shifted to online and modular learning. The Philippines transitioned to a new normal type of education, and constant innovations of teachers and active participation from other stakeholders are the driving forces behind its success. The Government Education Sectors developed Distance Learning with the aid of electronic modules and printed modules to ensure educational continuity and for each school to continue to fulfill its objective and vision of providing quality education to every Filipino learner.

Distance Learning refers to a learning delivery modality, where learning takes place between the teacher and the learners who are geographically remote during instruction. This modality has two different types: Electronic modules and Printed Modules Instruction (Quinones, 2020).

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Electronic modules are learning materials that can be accessed through the internet or can be saved in a Flash drive or Compact disk, which enables the students to access the learning materials using a laptop, computer, phone, or tablet. The e-copy of the learning modules like interactive and inclusive e-books, courseware, and other online and offline content formats are provided.

Based on the findings of Wijaya (2019), it shows that electronic modules have a significant effect on students learning.

Based on the researcher's personal experience with the electronic modules, it is observed that this latest and creative learning material intends to deliver a conducive and convenient way of studying. Electronic modules have the ability to make learning instruction open for learners around the globe, wherever they are, as long as there is an internet connection, learning is made possible, which subsequently plays a radical effect on the teaching and learning process.

On the other hand, printed modules are learning materials delivered to learners' homes or picked up by their parents or guardians at designated places within coordinated schedules. Printed modules refer to learning packets such as worksheets, activity sheets, and self-learning materials.

Based on the survey findings conducted by the Department of Education as cited by Bernardo (2020), Printed modules are the most popular type of Distance Learning. In the Philippines, all public schools currently use this learning modality. According to a survey, learning through printed modules emerged as the most preferred distance learning method for parents with children enrolled this academic year. This is also in consideration of the learners in rural areas where the internet is not accessible for online learning.

Research Performance can be defined as the outcome of the process of understanding the Research subject; it is the degree to which set standards and objectives in Research have been met. It is usually can be measured with Summative or formative assessments.

The main objective of this study is to determine if electronic modules affect the Research performance of the students compared to the printed type of modules.

(11) Materials and Methods or Methodology

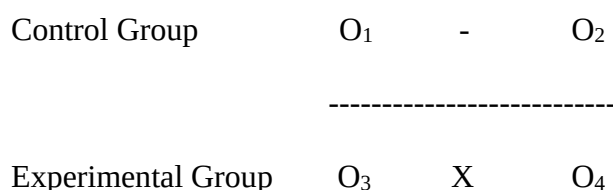
This True-experimental research used the pretest-posttest non-equivalent control group design. Conducted to investigate the effect of electronic modules and printed modules on the Research performance of Third-Year Industrial Technology Students during the school year 2021-2022. Data needed in this study made use of the 50-item Research Performance Test. Two learning materials were tried in this study, the use of electronic modules and the printed modules.

This study utilized the true-experimental method in determining the effect of the teaching materials on the students' performance in Research using the Pretest-Posttest nonequivalent control group design. According to Best and Khan, as cited in Joan (2013), True-experimental design resembled the quasi experimental design; however, the difference is it uses random assignment of the participants either in the control or experimental group.

Classes in the Industrial Technology department usually exist as intact groups and, teachers do not normally allow the classes to be dismantled and reconstituted for research purposes. This design has an advantage over others since it controls the major threats to internal validity except those associated with interaction and instrumentation (Fraenkel, Wallen & Hyun, 2012).

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In this study, the students were assigned to the control and experimental groups to control for selections and interaction. Were O_1 and O_3 were the pretest; O_2 and O_4 were the posttests; X was the treatment where students were taught using the electronic modules. The dotted line implies the involvement of intact groups. Group I was the control group, which received a pretest followed by the controlled condition and then the posttest. Group II was the experimental group that also received the pretest, treatment X, and the post-test. This research paradigm is represented in Figure 1.



This diagram shows that O_1 and O_3 were the observation results in the administration of pretest of both control and experimental groups. This was done before the conduct of the experiment or classroom instruction in the first grading period. It further shows that O_2 and O_4 are the observation results in the post-testing of the two groups. The results were recorded, tabulated, and analyzed using appropriate statistical tools.

EXPERIMENTAL PROCEDURES

There were three (3) experimental phases conducted by the researcher in this study: the pre-investigation phase, the investigation proper phase, and the post-investigation phase.

Pre-Experiment Phase - The researcher prepared the list of topics, syllabus, test items for pretest and post-test, and other materials needed in delivering the lesson before the actual experiment was carried out. The topics covered are Nature of research, Qualities of a Good Researcher and its Characteristics, Why research is of value, and Types of Research. The finished test items were presented to the panel of experts for face validation. Their suggestions and recommendations for the improvement of the instrument were incorporated into the revision of the same instrument. After the test items was face validated, the researcher sought permission to conduct the study from the Campus Administrator of Capiz State University-Main Campus.

Upon approval, the researcher conducted a pilot testing of the test instrument in an intact section in 3rd year Industrial Technology students of Capiz State University-Main Campus to determine the discrimination and difficulty index of the instrument. Test items that are too easy or too difficult were eliminated and was not included in the final draft of the test instrument.

After the reliability testing, the final draft of the questionnaire was sent to the target participants, which are the 3rd year Food Technology 3A of Capiz State University-Main Campus, and subjected to pretesting for both experimental and control groups, and this was done by conducting a synchronous class and answering the Research Performance Test; To eliminate the possible threat to the validity of the data the researcher requested the participants to open their cameras in the entire period and refrain from talking to anyone. After the pretest was conducted, the researcher retrieved the instrument, which was checked and tallied the results for analysis using the appropriate statistical tools, more so, the test materials were kept

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to be used at the end of the experimental phase.

Thereafter, the researcher did the grouping selection procedure; this was be done by picking up the rolled paper in a fishbowl with the names of the students written on it. The first student that was picked was assigned to the experimental group which was taught using a synchronous class and aided with treatment X which is the electronic modules. The second student that was picked was assigned to the control group that was taught using a synchronous class and the controlled condition, which is the printed modules, and the same procedure was done for the remaining students.

Experimental Phase - The experiment phase started on May 23, 2022, and ended on June 17, 2022. During this phase, the experimental group was taught using a synchronous class and aided with electronic modules, while the control group was exposed to a synchronous class with the support of printed modules.

During the conduct of the experiment, the two groups consisting of thirty students was exposed to the same classroom condition, which is google meet, an online website that was used to facilitate the discussion. Both the groups was taught by the same teacher to eliminate the idea of the “teacher factor”. The experimental group, which is composed of fifteen students, and aided with the electronic modules, while the control group consisted of fifteen students which used the printed modules.

Post-Experiment Phase - The researcher conducted the post-test for the control and experimental groups; this was done by answering the examination using their online learning tool, and to ensure that there would be no threat to the validity of the data, the students were asked to join the synchronous class through google meet while their cameras are all open to ensure they are the ones who are answering the questions.

After conducting the post-test, the instruments were retrieved, checked, and recorded. These were submitted to the statistician to be computed and analyzed. Both the pretest and posttest results were used for processing using the Statistical Package for the Social Science (SPSS) software. The differences in the pretest and post-test results was determined by submitting the data to statistical analyses using inferential statistical tools to answer the statement of the problem, and the hypotheses advanced in this study. The analyses of the data was placed in tabular form and textual presentation of the analysis and interpretation

(12) Results and Discussions

1. Research Performance in the Pretest and Posttest of the Control and Experimental Group Design

Table 1 presents the Research performance in the pretest and posttest of the experimental group using the electronic modules and the printed modules in the control group.

The pretest in the experimental group using the electronic modules ($M = 28.00$, $SD = 3.82$) is fair, while the pretest in the control group using the printed modules ($M = 26.80$, $SD = 3.95$) is fair. Moreover, in the post-test of the experimental group, the result is very high ($M = 44.67$, $SD = 3.11$) with a mean gain of 16.67, whereas in the post-test of the control group the result is high ($M = 30.53$, $SD = 5.83$) with a mean gain of 3.73. This shows that there is an increase in the Research performance of the students using both approaches in teaching.

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Pretest					Posttest				
Group	Mean	Desc.	SD	N	Mean	Desc.	SD	N	Mean Gain
Experimental	28.00	Fair	3.82	15	44.67	Very High	3.11	15	6.36
Control	26.80	Fair	3.95	15	30.53	High	5.83	15	3.33

Scale	Description
40.01 – 50.00	Very High
30.01 – 40.00	High
20.01 – 30.00	Fair
10.01 – 20.00	Low
0.00 – 10.00	Very Low

The “fair” Research performance of the students in the pretests of experimental and control groups showed that Third-Year Industrial Technology Students have fair enough knowledge and skills in Research and they can manage to have acceptable performance in a Research test. This is because the Research subject is also taught in Senior High School for both grade 11 and grade 12 so they already have a background about the subject. Thus, the students were able to adapt and develop their Research skills out of this.

The result of the post-test revealed that learning had taken place between the two groups. Both groups had improved their Research performance from “fair” to “Very High” in the experimental group and from “fair” to “high” in the control group. It is also important to notice that students in the experimental group using the electronic modules had greater mean gain than the control group which used the printed modules. This implies that Third-Year Industrial Technology Students prefer to use electronic modules over printed modules.

The finding of the study affirms the conclusion of Arani (2014) which states that the use of online learning resources is more effective than the traditional methods.

The result also substantiates the findings of Jaenudin (2017) that the existing problem in conventional learning, utilizes less learning media. Therefore, the effectiveness of learning is determined by how far the teacher conducted the readiness of planning. Media utilization in learning can be a tool and help the teacher in the learning process.

The result likewise agrees with the conclusions of Sanaky (2013) that using electronic modules aids the learning process which stimulates thought, attention, and students' willingness to enhance the effectiveness and efficiency in achieving learning objectives.

The result supports the findings of Wijaya (2019), which revealed that the average scores in the pre-test and post-test of the students differ descriptively after the electronic modules intervention.

2. Differences in the Research Performance in the Pretest of the Control and Experimental Group

Table 2 presents the differences in the pretest scores of the experimental and control groups.

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Data show statistically that there is no significant difference $t(0.85^{ns}) = .405$, $p > .05$ in the pretest scores on the Research performance both for experimental and control groups. This indicates that the experimental and control groups are comparable in their Research performance.

Table 2

The t-test in the Pretest Scores of the Experimental and Control Groups

Group	Mean	SD	t	df	Sig
Control	26.80	3.95	0.85 ^{ns}	28	0.405
Experimental	28.00	3.82			

* $p < 0.05$ significant @5% alpha level

^{ns} $p > 0.05$ significant @5% alpha level

The no significant difference in the pretest scores of the students in the Research performance test revealed that the students who use the electronic modules and printed modules may have the same performance in Research. This shows that the experimental and control groups are similar and comparable to each other. The results also imply and validated the grouping selection procedure that the participants were equally matched.

The result can also be linked to the theoretical framework of the present study, the Theory of Cognitive Load of Sweller (2011), which revealed that since both of the groups have collective similarity, they have similar loads in their working memory or equally preconceived knowledge and they had an equal chance to absorb new loads or new concepts, and subsequently both of them can have an analogous Research performance.

The result of this study conforms to the result of, the effect of Online-Modular Instruction by Valderama (2012) which showed that the students' mean cumulative scores in the pre-test for the experimental and control groups were found to have no significant difference from each other. This further revealed that the two groups of students have equal performance before the experiment.

Moreover, the findings of this study affirm the contention of Burk (2011) in Online Learning Indicators, further revealed that there was no significant difference in the amount of prior knowledge among students treated under the control and experimental group based on the pre-test mean scores which implied that the two groups of respondents have the same amount of prior knowledge on the topics before the conduct of the teaching methodologies.

3. Differences in the Research Performance between the Pretest and Posttest of the Control and Experimental Groups

Table 3 presents the differences in the pretest and posttest scores of the control and experimental groups.

Data show that there is a significant differences $t(6.014^*) = 0.000$, $p < 0.05$ in the pretest and post-test scores in the control group and $t(14.248^*) = 0.000$, $p < 0.05$ in the pretest and post-test scores in the experimental group on the Research performance in favor of the experimental group using the electronic modules. This means that students using electronic

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modules have a better Research performance than the students using the printed modules

Table 3

The t-test in the Posttest Scores of the Experimental and Control Groups

Group	Test	Mean	SD	t	df	Sig
Control	Pretest	26.80	3.95	6.014*	14	0.000
	Posttest	30.53	5.83			
Experimental	Pretest	28.00	3.82	14.248*	14	0.000
	Posttest	44.67	3.11			

* $p < 0.05$ significant @5% alpha level

The significant difference that exists in the pretest and posttest scores of the experimental group using the experimental treatment which is the electronic modules, suggests that utilization of treatment X is an effective tool for improving Third-Year Industrial Technology Students' Research performance. Comparably, its counterpart which is the Printed modules also helps to improve the Research performance of the students since the result shows that there is a significant difference in the pretest and posttest results of the participants.

The result further revealed that the use of electronic modules and printed modules have contributed to improving the Research performance of students; It is because both are effective teaching tools in this dire pandemic situation since they can still be a source of information in the absence of face-to-face classes which is facilitated by a teacher. Also, learners are both active participants in the learning process since they still have to read and understand their lesson but just only placed in a different format.

In addition, both of the treatments had a significant effect on the development of Research Performance. It is due to the effectiveness of both learning materials provided to the students for the teaching and learning of the theoretical aspects of a topic, which consequently resulted in an increase in the performance among the users.

Alternatively, both of the teaching tools used in teaching practical skills could have helped to increase the theoretical knowledge acquired by the learners but to comparable levels among the two treatment groups. Thus, the two instructional materials were pedagogically equivalent in terms of theoretical knowledge acquired.

The finding of this study is in conjunction with the Cognitive Load Theory (Sweller, 2011), which suggests that learners can absorb and retain information effectively only if it is provided so that it does not "overload" their mental capacity. In other words, rather than an infinite supply of data, short-term memory, or working memory, can only hold a limited quantity of information at a time. Electronic modules contain images, video, and simulation tools that help explain a particular concept easily, which will not bombard or overload the students' mental capacity. On the other hand, printed modules are highly contextualized learning packets, consolidating and summarizing the most essential ideas needed by the students to understand a topic.

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It is also emphasized that there is a significant increase in the Research performance of students using the electronic modules over the printed modules; It is because aside from the presence of the teachers monitoring the activities by means of synchronous class, technology is used to enrich the learning experiences of students through images, videos and simulation tools that are presented in the electronic modules.

The finding of this study strengthens the conclusion of Suarsana et.al (2013) who states that media is used as aids by the teacher as the evidence of classroom effectiveness and substitutional media in the learning media system.

The result reinforces the findings conducted by Istuningsih et al. (2018) that have proven the use of electronic modules is very effective in improving students' learning outcomes since electronic modules is a combination of technology and teaching materials that can be appealing to the students as it contains multimedia elements such as audio, video, and images inside which can help to understand the materials so that this media can create more interesting learning.

On the other hand, the result contradicts the findings of Sejpal (2013) that the utilization of printed modules takes into account the individual differences of the students and enables them to study at their own pace. It contains well-defined objectives that cover every single element of the topic or the entire unit consisting of various discussions and activities that develop their cognitive and affective skills through self-learning.

4. Differences in the Posttest Scores of the Control and Experimental Groups

Table 5 presents the differences in the posttest scores of the control and experimental groups.

Data show that there is a significant difference $t(8.285^*) = 0.000, p < 0.05$ in the posttest scores on the Research performance in favor of the experimental group ($M = 44.67$; $SD = 3.11$) using the electronic modules.

Table 5

The t-test in the Posttest Scores of the Experimental and Control Groups

Group	Mean	SD	t	df	Sig
Control	30.53	5.83	8.285*	28	0.000
Experimental	44.67	3.11			

* $p < 0.05$ significant @5% alpha level

The significant difference in the posttest scores of the experimental and control groups implies that the students who use the electronic modules have a better score than the students who use the printed modules; this also means that the intervention that the experimental group received can lead to a better Research performance than the controlled treatment.

The significant difference in the post-test scores is due because electronic modules are more attractive than printed modules because it has several multimedia elements that make learning more fun and interactive and it is an effective learning resource that is used as a

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medium to stimulate the process of thoroughly understanding the topic. Electronic modules allow the students to access the materials and study anywhere at any time as long as they have their devices and connectivity with them.

The result reinforces the previous research conducted by Lim & Spahat (2016) that the electronic modules are a combination of technology and teaching materials that can appeal to students because there are multimedia elements (audio, video, and images) inside which can help to understand the materials so that this media can create more interesting learning.

The study supports the findings of Trilestari and Almunawaroh (2020), which showed that electronic modules have a significant effect on the students' Research performance on the post-test scores of the experimental group using the electronic modules; It revealed that the learners enjoyed the e-resources and make their learning more effective.

However, the results of this study contradict the findings of Sadiq and Zamir, (2014), which stated that Printed modules are better than electronic modules since it contains a set of learning opportunities organized around a well-defined topic that contains the elements of ordinate dictation, categorical objectives, edifying cognition activities, and evaluation utilizing criterion-referenced measures.

5. Effect Size of Electronic modules on the Research Performance

The effect size of electronic modules on the Research performance is shown in Table 6.

Data show that statistically, electronic modules have a large effect (44.67%) on the Research performance of students (*Cohen's d = 3.03*). This means that using electronic modules can greatly help improve the Research performance of students.

Table 6

Cohens' d on the Effect of electronic modules on the Research performance

Group	Mean	SD	Cohen's d	Remark
Control	30.53	5.83	3.03	Large
Experimental	44.67	3.11		

Scale	Description
0.2	Small
0.5	Medium
0.8	Large

The result implies that there is a "Large" improvement in the Research performance of the students using the electronic modules over the printed modules. It is because there are certain multimedia aspects present in electronic modules that appear to be more interesting to the students such as audio, video, and simulation tools. Unlike in the printed modules, the students may find it less engaging since the materials are static which is not engaging for teaching the subject.

This implies that electronic modules can help improve the students' Research performance; It is because the students can maximize their learning potential with the help of electronic modules since it contains some technological tools such as images, videos, and simulation tools that would make their learning process easier and comprehensible.

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Moreover, in accordance with the accepted principle in teaching and the philosophical idea of Dale's cone of experience that seeing images, videos and participating in a hands-on workshop through simulation tools can greatly help to remember what they've learned in the class and will result in a better learning outcome when more senses are used; Also according to John Sweller's theory of Cognitive load, these technologies don't overload the mental capacity the students resulting to a better understanding of the lesson.

The result is in agreement with the conclusions of DeWitt et. Al (2014) that electronic modules utilization that combines technology development and learning can provide a new learning experience in building students' knowledge and using electronic modules in learning is effective to give interaction insights and develop students' knowledge.

The result conforms to the findings of Abalos, Santiago & Santos (2012), that there was a significant positive effect in using the electronic modules, which revealed that electronic modules have a positive actual effect on the students' scores because electronic modules cater to different learning styles as well as it allows more students to become interested in the learning process due to its interactive features.

The result argues the statement of Gömleksiz (2018), which showed that when the students accept the printed modules it has been shown that it will increase their readiness to learn Research. The figures and pictures used in the learning packages helped them learn Research a lot. They express that the exercises used in learning packages support both classroom and module activities. The evaluation questions have been found efficient. The students remarked that the exercises in the modules are in line with the content of the modules and the objectives were stated clearly and understandable.

(13) Conclusions and Recommendations

CONCLUSIONS

The following conclusions were drawn based on the study's major findings as indicated above.

1. Given that both groups have "fair" Research performance in the pretests it is reasonable to think that both groups have a collective similarity. It also indicates that students have a fairly enough Research foundation since the subject is being taught already in Senior High School for both grade 11 and grade 12.

However, the "fair" pretest result also indicates that Third-Year Industrial Technology Students' Research performance should be improved since the students have not reached the high scores and there are still some areas where their Research abilities need to be developed.

The improvement in students' Research performance from "Fair" to "High" in the control group and from "Fair" to "Very High" in the experimental group reveals that there is an effective transfer of learning among students using both learning resources.

Students can still improve their Research performance when using the appropriate modality supplemented with a teacher-mediated synchronous class. One key factor to take into account is knowing our students who will be using the modality since some students do not even have stable internet access, which is a significant issue for many students not just in our area but also in other parts of the Philippines.

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Though both data show that the students' Research performance has improved, it is apparent that the experimental group, which used electronic modules, has shown a greater improvement. It is possible to conclude that using this learning tool will improve students' Research performance more effectively than the traditional treatment using the printed modules. As a response, teachers and school administrators should consider implementing electronic modules in their respective institutions.

2. The Research performance of Third-Year Industrial Technology Students using the electronic modules and printed modules do not differ significantly. This result indicates that even without any intervention the students could manage to have a fair enough score to pass the Research Performance Test, but the pre-conceptions of the topics were insufficient and thus, require the teaching and learning process of more concepts to be learned.

New Concepts can be introduced to the students so they can grasp especially when they learn them through the aid of proper instructional material. The prior knowledge, whether correct or not, helps the students master new content because they still have spaces to be filled and their mental capacity still does not overload with accordance to Cognitive Load theory the students still have the opportunity to learn continuously.

In conjunction with the grouping selection procedure, the researcher made sure that the participants were equally situated to one another before the intervention was introduced to the two groups, so there would be no biased selection, as a result, the data shows both the groups were just comparatively equivalent.

The students' similar backgrounds, such as major, age level, and year level, were taken into account, which leads to the notion of a seemingly similar level of Research proficiency. Thus, it is viable that students in the experimental and control groups will have an equal chance of improving their Research performance.

3. There is a significant difference in the Research performance of Third-Year Industrial Technology Students in the pretest and posttest of the control and experimental group. The result signifies that both of the treatments were effective in improving the Research performance of the students.

The significant difference in the pretest and posttest scores also indicates that utilization of electronic modules and printed modules should be continued as an alternative to the traditional method, especially in this dire situation since the electronic modules offer a new experience in studying, it opens up the portal of knowledge where students can discover new ideas, exploring other dimensions that remain to be unfathomed through cybernated activities. Into the bargain, teachers are advised to utilize some tools that support Augmented Reality (AR) wherein students can utilize to develop their deeper understanding of the topic through 3d images which helps them to have a vivid and convincing learning experience.

It is also suggested to use the interactive features of electronic modules such as pen tool, annotation, search option, page zoom, and read aloud, all of which assist students in taking notes or highlighting the essential information in the module.

In winding up, is also advisable to continue using the printed modules where it can also be utilized as a learning tool since it contained only the necessary information that the students need in order to understand their lesson in Research. The printed modules also contain highly contextualized lessons where the topics are integrated into the actual application of the set competencies.

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4. The significant difference in the posttest scores of the control and experimental groups signifies the use of e electronic modules is highly effective in improving the Research performance of the students, thus, is advisable to utilize this learning tool to aid the class.

It is also implied that institutions may also utilize electronic modules in their respective school since it could be one of the solutions to support the students who are studying at home due to the life-threatening Corona Virus outbreak because electronic modules provides digital teaching and learning materials and it's compiled with interactive videos, audios, pictures and simulation tools to improve students interest in learning. It also can be accessed anywhere at any time as long as they have internet connectivity. To cover the problems of data limit, and intermittent connection, the students are advised to download or take a screenshot of the materials ahead of time so they can even access the learning materials offline.

5. The result indicates that electronic modules have a large effect on the Research performance of students; This points out to use this beyond compare learning tool which can be an effective way to develop the students' Research performance. This is due to the fact that electronic modules offer better content since it contains some interactive features such as images, videos, simulation tools, 3d tools, annotation tools, an aesthetically pleasing graphical interface, very user friendly, which consequently motivates the students to engage in the process of learning acquisition.

Over and above that, the large effect substantiates the notion to utilize the electronic modules since the students in the treatment group who uses the electronic modules did better in acquiring new information because they promote long term retention and help the students understand complex ideas easier since they are exposed to these avant-garde learning materials. The contents in the electronic modules are really engaging for the students owing to the fact that they are more realistic and graspable, as it has the ability to create a learning experience more personalized to the students, because students can have a bare hand experience in doing some simulation tasks, thus, electronic modules can be really utilized as an effective learning tool in the classroom.

RECOMMENDATION

In view of the aforementioned conclusions that have been formulated, the researcher offers the following recommendations:

1. Students are advised to download or take a screenshot of the learning modules ahead of time, especially if they are experiencing intermittent internet connectivity, in order to minimize technical difficulties that could lead to distractions and a poor learning experience.

Students are also recommended to have a backup internet connection, such as mobile data especially if they are taking quizzes or major examinations because there are some instances that their internet connection might be lost so if this technical difficulty will transpire, they have backup connectivity to continue answering their examinations.

Students are also advised, particularly those who are not digitally inclined and are unable to use their computers efficiently, to seek assistance from their teachers, parents, or guardians if they are experiencing difficulties so that they can address and overcome those concerns. Basic computer literacy courses will improve students' computer expertise; Being compelled to have a basic understanding of the computer hardware and software will allow

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them to join an online class without interruptions.

2. Teachers are recommended to thoroughly explore the electronic modules because these will enable them to understand the unique and helpful features of these learning materials such as grade book, auto-scoring of quizzes and examination, simulation activities, 3d tools, chat, forum, and e-library; These features will help them to minimize their work and improve their efficiency. Also, these tools will help them to save their time just like the very time-consuming test checking, and grading.

It is recommended to utilize these unique features to lessen their job; for instance, on the use of auto-scoring of quizzes and examination, the teachers must set up their test beforehand and input the correct answers in the program so after the students finish taking the quiz or examination they will know their scores automatically and will automatically encode the score in the grade book or commonly known as the class record. In this account, they would tremendously save their time in checking and recording.

Teachers are recommended to have refresher training on how to use the electronic modules. It should become a standard practice to have annual training on how to use the platform, this would be very beneficial to the newly hired teachers in understanding the different features of the electronic modules. It would also serve as a review to the tenured teachers, allowing them to go over again or learn some new portal features.

It is advised that teachers must update their instructional materials every time a new school year begins. It is recommended that the content must be updated on a frequent basis because our world is constantly changing and new knowledge may become available that is more relevant, relatable, and engaging to the students.

The teachers may evaluate and update the quality of the layout, transitions, font style, appropriateness of the video, pictures, as well as the balance between visuals and text, as they play a salient role in the teaching and learning processes. The overall functionality, graphical interface, interactivity, and user-friendliness must also be evaluated and updated. Correspondingly, could promote collaborative and interactive pedagogical methods and subsequently enhance the quality and improve the students' performance.

Moreover, Teachers are recommended to have constant communication with the students for monitoring purposes. Teachers must open their line, especially during working hours to monitor their students since there are some instances that students are experiencing some technical difficulties or they don't understand some parts of their modules and none can help them at home or while their parents are at work, so teachers should keep their lines open to fix and address their problems.

3. The school administrators may support this by providing rolling in equipment and facilities. They may provide more and up to speed servers to accommodate all the students accessing the server simultaneously without experiencing any delay or lag since it might affect the students' accessibility and connectivity to the server.

The school administrators may conduct training and seminars for the teachers that could help them increase their knowledge on the electronic modules and made them aware of the new features of the learning materials. This can be done by continuing and doing the annual training for teachers for them to be able to update the lesson materials and run through the old and new features of the electronic modules so they will be able to develop their abilities and continuously adapt to emerging innovation of this state of the art learning materials

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4. Parents are advised to constantly monitor their child's performance at home. Due to the Coronavirus outbreak, the students are advised to stay at home to contain the spread of the disease-causing virus. As an alternative to the conventional classroom, the Government advised the different institutions to utilize distance learning and other modalities that would serve as a replacement to the traditional classroom setup. In to this point, the parents are recommended to keep an eye on their child, since there may be some times when they might ask for assistance to resolve connectivity issues or difficulties understanding their lesson modules.

Parents are also recommended to support their children's needs while they are exposed to this type of learning modality. Parents can help their children by providing an up and running device and reliable internet connection so they can attend their class or can study without any interruptions.

(14) Output

In light of the current dire pandemic situation, in which many institutions, including us here at CAPSU, are compelled to use modular instruction, this true-experimental research study bridged the existing gap, provided an in-depth understanding of the phenomenon and offers a solution that can maximize the learning experience of the student. As a result, this study was able to determine which one can truly provide a better Research performance.

Providing "Dekalidad nga Serbisyo" is what we strive for here at CAPSU; the aforementioned quotation is a perfect representation of our values and ideals, and as such, finding the most useful educational resource that will assist us in achieving this objective is our top priority.

(15) Technology Transfer Activity Conducted

The findings of this study will be able to assist the adopted community in maximizing their learning potential through the use of the learning resource that this research found out to be effective, since it contains interactive tools such as images, videos, and simulation tools that will make their learning process easier and more understandable.

The proponent of this study hopes that the Industrial Technology department will continue its community outreach program teaching members of the community about HVAC-RT, Electrical, Food processing and preparing, and dress making. However, rather than teaching them with the typical learning resource, the proponent hopes that the electronic modules will be integrated since the present study has proven that it is an effective learning resource that will maximize the learning potential of the participants.

(16) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
- Research Assistant		5,000.00
- Statistician		5,000.00
- English Editor's Fee		5,000.00

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B. Maintenance and Other Operating Expenses (MOOE)		
- Meal allowance during data gathering		10,000.00
- Miscellaneous (for incidental expenses)		5,000.00
- Printer		7,000.00
- Printer Ink assorted colors		2,370.00
- Bond paper substance 20 8x11 ½		930.00
- Publication Fee		20,000.00
- Presentation		20,000.00
Total		80,300.00

(17) Acknowledgments

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